

1. Dimensional analysis

↳ Primary dimensions, dimensions not formed from other dimensions

MLT:	Typically this	FLT:	
[Mass] = M		[Force] = F	In MLT, $F = \frac{ML}{T^2}$
[Length] = L		or [Length] = L	
[Time] = T		[Time] = T	

In heat transfer, [temp] = θ

In electric system, [electric current] = I

↳ Secondary dimensions, derived by combining primary dimensions

ex. $[p] = \frac{M}{L^2}$

or $= \frac{F}{L^2}$ in FLT

↳ Buckingham Pi Theorem

$\pi \rightarrow$ dimensionless number, ex. L/D

(# of π terms) = (# of variables) - (minimum required # of reference dimensions)

ex. $A = f_1(B, C, D, E, F)$,

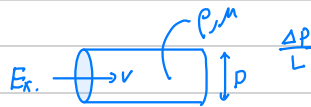
of π terms = $6 - 3 = 3$

typically primary dim $\rightarrow$ Not always, ex.

$[A] = \frac{M}{L^3}$	primary dim $\Rightarrow M, L, T$
$[B] = \frac{M}{L^3 T^2}$	
$[C] = \frac{M T}{L^3}$	

net dim $\Rightarrow \frac{M}{L^3}, T$

↳ Method of repeating variables:



1.) List all independent variables

$\frac{\Delta P}{L} = f_1(v, D, \rho, \mu)$

2.) Express each variable in terms of primary dimensions

$[\frac{\Delta P}{L}] = \frac{kg \cdot m}{s^2 \cdot m^2} = \frac{M}{L^2 T^2}$, $[v] = \frac{L}{T}$, $[D] = L$, $[p] = \frac{M}{L^3}$, $\mu = \frac{M}{L T}$

3.) Determine the required number of pi terms using Buckingham Pi theorem

= $5 - 3 = 2$

4.) Choose some repeating variables, s.t. # repeating variables = # of reference dim

Rules: a. Don't select the dependent variable, ex. $A = f(B, C, D, E, F)$ ✗

b. All ref dimensions must be represented in the repeating group, ex. If ref = MLT, $B = [M]$, $C = [M^2]$, $D = [T]$ would not work

c. Each repeating variable that we choose should be dimensionally independent of others.

Choose ρ, v, D

5.) Form a π term by multiplying one non-repeating variable by the product of repeating variables, each raised to an exponent, carrying combination $\rightarrow$ dimensionless.

$\pi_1 = \frac{\Delta P}{L} v^a D^b \rho^c$

$1 = [\frac{M}{L^2 T^2}] [\frac{L}{T}]^a [L]^b [\frac{M}{L^3}]^c$

$M: 1 + c = 0 \rightarrow c = -1$
 $T: -2 - a = 0 \rightarrow a = -2$
 $L: -2 + a + b - 3c = 0 \rightarrow b = 3c + 2 - a = -3 + 2 + 2 = 1$

$\therefore \pi_1 = \frac{\Delta P}{L} v^{-2} D^1 \rho^{-1}$

$$\pi_2 = \mu V^a D^b \rho^c$$

$$1 = \left[\frac{\mu}{LT} \right] \left[\frac{L}{T} \right]^a \left[L \right]^b \left[\frac{M}{L^3} \right]^c$$

$$M: 1 + c = 0 \rightarrow c = -1$$

$$T: -1 - a = 0 \rightarrow a = -1$$

$$L: -1 + a + b - 3c = 0 \rightarrow b = 3c + 1 - a = -1$$

$$\pi_2 = \mu V^{-1} D^{-1} \rho^{-1}$$

6) Check that all resulting π terms are dimensionless

7.) Express new relationship among π terms

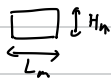
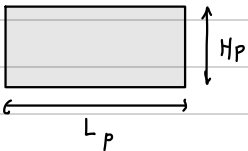
$$\pi_1 = f_2(\pi_2)$$

$$\frac{\Delta p D}{\mu V} = f_2 \left(\frac{\mu}{V D \rho} \right)$$

↳ Modeling & Similarity

- Types of similarity

1.) Geometric similarity: all lengths are scaled by the same factor



$$\frac{L_p}{L_m} = \frac{H_p}{H_m} = \text{const}$$

2.) Dynamic similarity $\Rightarrow$ corresponding forces scale by a constant factor

3.) Kinetic similarity $\Rightarrow$ velocities at corresponding points scale

↳ Principle of Similarity

- Ensure similarity between model & a prototype, geometric similarity must exist and all independent π 's must be identical

Then the solution in dimensionless form are guaranteed to be equal

↓

If geometric similarity & $\pi_{2,m} = \pi_{2,p}, \pi_{3,m} = \pi_{3,p} \dots$ then

$$\pi_{1,m} = \pi_{1,p}$$

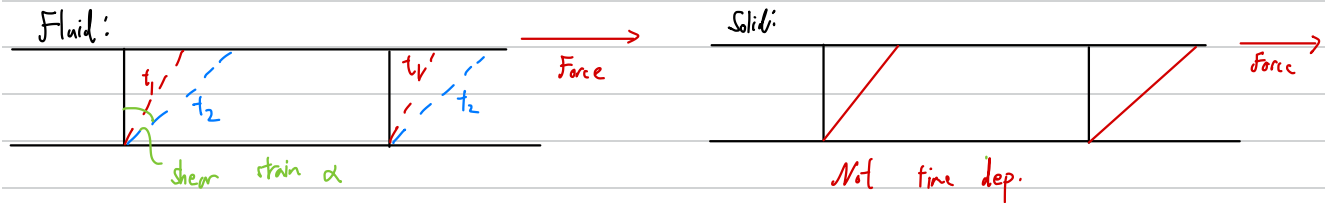
↳ Reynolds number, $Re = \frac{\rho V L}{\mu}$

2 Introduction to Fluid Mechanics

↳ Definition of fluids:

A substance that deforms continuously under the application of a shear (tangential) force

No slip condition: 0 relative velocity b/w fluid and boundary



↳ Statistical vs Continuum Approach

Stats:

- Uses probability and statistics in macroscopic behaviour
- Kinetic theory in molecular viewpoint

Continuum (This course)

- Limit and differential concept, let $L, A, V \rightarrow 0$
ex. density at point C,

$$\rho = \lim_{\delta V \rightarrow 0} \frac{\delta m}{\delta V}$$

known as intermediate region, where $\delta m / \delta V$ is const.

- Knudsen number, Kn (unitless):

$$Kn = \frac{\text{microscopic length scale}}{\text{macroscopic length scale}}$$

- Continuum holds, $Kn < 0.01$
- Continuum fails, $Kn > 0.01$
 - o $0.01 < Kn < 0.1$, Slip flow, No slip condition does not hold, continuous substance
 - o $0.1 < Kn < 3.1$, Transitional flow, Fluid is not a continuum, molecules still interact to a considerable degree
 - o $3.0 < Kn$, Free Molecular flow, Far apart molecules, ~ 0 interaction, modelled as collection of non-interacting impacts
- Examples of continuum failing:
 - ↳ Spacecraft entering the earth's atmosphere, microscopic $\gg$
 - ↳ Nanotech devices, microscopic $\approx$ macroscopic
 - ↳ Granular flows, microscopic $\approx$ macroscopic
 - ↳ Shock waves, microscopic $\ll$

↳ Properties

- Density [kg/m^3], mass per unit volume,

$$\rho = \frac{m}{V}$$

- o In liquid, constant for temperature; In gas, changes with temperature, pressure
- o Constant density for all conditions $\rightarrow$ incompressible.

- Specific gravity [none], ratio of liquid density to water density at same condition, typically 4°C

$$\text{SG} = \rho_{\text{liquid}} / \rho_{\text{H}_2\text{O}}$$

- Specific weight $[\text{N/m}^3]$, weight of substance per unit volume

$$\gamma = W/V = \frac{\rho V g}{V} = \rho g$$

- Specific volume $[\text{m}^3/\text{kg}]$, volume per unit mass

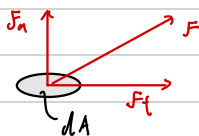
$$\bar{v} = \frac{1}{\rho}$$

↳ Body forces

- Forces w/o physical contact
- Proportional to mass of fluid
- Gravitational field:
 $d\vec{F}_g = \rho dV \vec{g}$
- Magnetic fields.

↳ Surface Force

- Forces generated at the surface of fluid element, by direct contact with other particles or solid surface
- Divided into normal, tangential components



- Surface forces leads to stresses

↳ Stress $[\text{N/m}^2, \text{kg/ms}, \text{Pa}]$, force per unit area.

- Separated into normal / tangential

★ Hydrostatics (fluids at rest) $\longrightarrow$ shear stress $= 0$, pressure = normal stress

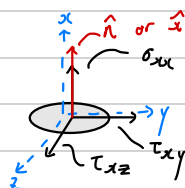
↳ Stress fields:

Normal stress, $\sigma = \lim_{\delta A \rightarrow 0} \frac{\delta F_n}{\delta A}$

Shear stress, $\tau = \lim_{\delta A \rightarrow 0} \frac{\delta F_t}{\delta A}$

- Stress in Cartesian,

τ_{xy} $\xrightarrow{\text{direction of action}}$
 $\xrightarrow{\text{surface of application}}$



- Stress tensor

$$\begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_{yy} & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_{zz} \end{bmatrix}$$

- Sign depends on outward/inward normal chosen

↳ Viscosity

- Measure of liquids inertial resistance to deformation under shear stress.

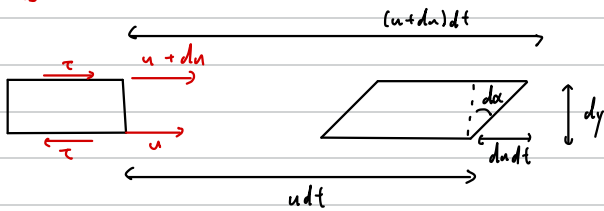
- Solids:

$$\tau = \gamma \Delta \alpha, \gamma \rightarrow \text{shear modulus here}$$

- Fluids:

$$\tau = \mu \times \frac{da}{dt} \quad (\text{in Newtonian Fluids})$$

shear stress
viscosity
strain rate



$$\tan da = \frac{du dt}{dy}$$

$$\tan da \approx da$$

$$\frac{da}{dt} = \frac{du}{dy}$$

For **any** fluids:

For Newtonian fluids,

$$\tau = f\left(\frac{da}{dt}\right) = f\left(\frac{du}{dy}\right)$$

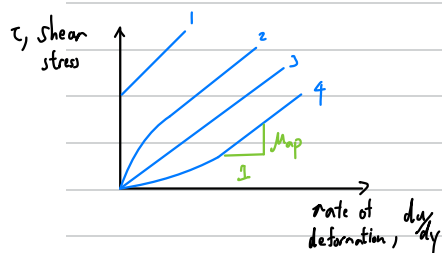
$$\tau = \mu \frac{da}{dt} = \mu \frac{du}{dy}$$

- Viscosity [$\text{N}\cdot\text{s}/\text{m}^2$, $\text{Pa}\cdot\text{s}$] or poise, $1 \text{ poise} = 0.1 \text{ N}\cdot\text{s}/\text{m}^2$

- Inviscid fluid $\rightarrow \mu = 0$ (superfluid He)

↳ Non-Newtonian Fluids

- Apparent viscosity, μ_{ap} , viscosity at local conditions



1) Bingham plastic, solid until minimum shear stress, linear relation b/w τ , $\frac{du}{dy}$ at higher τ

ex. drilling muds, toothpaste

2) Shear thinning, μ_{ap} decreases as $\frac{du}{dy}$ increases

ex. paints, blood, cookie dough

3) Newtonian

4) Shear thickening, μ_{ap} increases as $\frac{du}{dy}$ increases

ex. cornstarch-water mixture

↳ Temperature, Pressure on μ .

- Liquid: Temperature $\uparrow$, $\mu \downarrow$, more energy to overcome intermolecular force, more free movement

- Gases: Temperature $\uparrow$, $\mu \uparrow$, negligible intermolecular forces, more molecular collision $\rightarrow \uparrow$ viscosity

↳ Kinematic viscosity $[m^2/s]$, or $1 \text{ St} = cm^2/s$

↳ Compressibility

- Bulk modulus, E_v , change in pressure to change in volume.

$$E_v = - \frac{dp}{dV/V}, \text{ negative since } \uparrow P \rightarrow \downarrow V$$

$$E_v = \frac{dp}{d\rho/\rho} \quad \text{since} \quad m = \rho V$$
$$dm = d\rho V + \rho dV$$
$$\frac{dV}{V} = - \frac{d\rho}{\rho}$$

- Bulk modulus $\uparrow$, compressibility $\downarrow$

incompressible substance has $E_v \rightarrow \infty$

- Ideal gas:

$$p = \rho R T \rightarrow \left. \frac{dp}{d\rho} \right|_T = RT \rightarrow E_v = \left. \frac{dp}{d\rho} \right|_T = p = P$$

3. Hydrostatics

★ No shear force / stress for Hydrostatics

↳ Pressure



- Microscopically rough, Macroscopically flat surface
- Statistically, all tangential force $\Sigma = 0$, pressure acts **normal to surface**

- Pressure force: Molecule collides and exerts a force on the wall, changing molecule momentum, results in pressure force

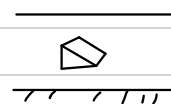
$$d\vec{F}_n = -P \cdot dA \cdot \hat{n}$$

outward pointing

↳ Pressure at a point:

Pascal's Law: Pressure indep. of direction $\rightarrow P_x = P_y = P_z = P_s$ any surface orientation

↳ for fluid element with accelerated motion, moving as a rigid body $\rightarrow$ i.e. no relative motion b/w particles, no shear



* for shear present, this is not necessarily true, P taken as average.

↳ Equation of pressure field.

$$-\vec{\nabla}P + \rho \vec{g} = \rho \vec{a}$$

not body force per unit volume

not total force per unit volume

not pressure force per unit volume

or when $\vec{g} = -g\hat{k} \rightarrow -\vec{\nabla}P - \rho g\hat{k} = \rho \vec{a}$

Assumptions:
① No shear

- Fluid at rest,

$$\vec{\nabla}P = \rho \vec{g}$$

Assumptions:

② $\frac{\partial P}{\partial x} = 0, \frac{\partial P}{\partial y} = 0$

$$\vec{\nabla}P = -\rho g\hat{k}$$

$$\frac{\partial P}{\partial z} = -\rho g$$

③ $\vec{a} = \vec{0}$

$$\frac{\partial P}{\partial z} = -\gamma$$

specific weight

- Incompressible Fluid at rest,

Assumptions:

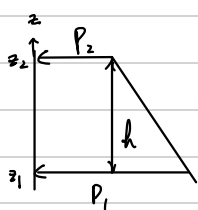
④ g const

⑤ ρ const

$$P_2 - P_1 = -\gamma(z_2 - z_1)$$

$$P_2 - P_1 = \rho g(z_1 - z_2) \quad \text{or}$$

$$P_1 - P_2 = \rho g(z_2 - z_1)$$



- Compressible Fluid at rest,

Assumptions:

ex. $P = \frac{P}{RT}$, isothermal ideal gas

no ④, ⑤

$$P_2 \int \frac{dP}{P} = - \int_{z_1}^{z_2} \frac{\rho g}{RT} dz$$

$$\ln\left(\frac{P_2}{P_1}\right) = \frac{g(z_1 - z_2)}{RT}$$

$$P_2 = P_1 e^{g(z_1 - z_2)/RT}$$

↳ Definitions of pressures

- Absolute pressure: Referencing vacuum

- Gauge pressure: Relative to local atmospheric pressure

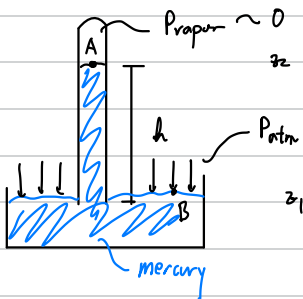
$$P_{abs} = P_{atm} + P_{gauge}$$

- Negative P_{gauge} (suction/vacuum pressure)

$$P_{vacuum} = -P_{gauge} = |P_{gauge}|$$

- Units lb/in^2 (psi) **Brits**, N/m^2 (Pa) **SI**

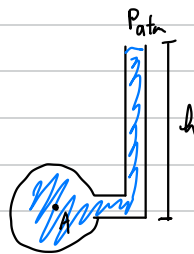
↳ Mercury Barometer



$$P_{atm} - P_{vapor} = \rho_{Hg} g h$$

$$P = \rho_{Hg} g h$$

↳ Piezometer

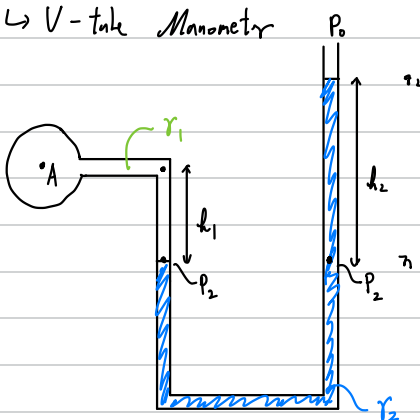


$$P_A - P_{atm} = \rho g h$$

$$P_A = P_{atm} + \rho g h$$

Using **gauge pressure**,
 $P_A = \rho g h$

↳ V-tube Manometer



$$P_2 - P_0 = \rho_2 g h_2$$

$$P_2 - P_A = \rho_1 g h_1$$

$$P_A = P_2 - \rho_1 g h_1$$

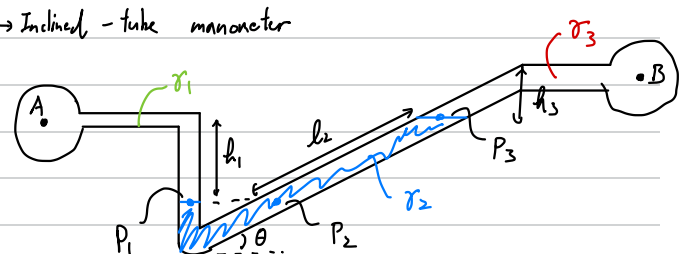
$$P_A = \rho_2 h_2 - \rho_1 h_1$$

If A contains gas,

$$\rho_1 h_1 \sim 0$$

$$P_A = \rho_2 h_2$$

↳ Inclined - tube manometer



$$P_1 - P_A = \rho_1 h_1$$

$$P_2 = P_1$$

$$P_2 - P_3 = \rho_2 h_2 \sin \theta$$

$$P_3 - P_B = \rho_3 h_3$$

$$P_2 - P_B = \rho_2 h_2 \sin \theta + \rho_3 h_3$$

$$P_2 - P_A = \rho_1 h_1$$

$$P_A - P_B = \rho_2 h_2 \sin \theta + \rho_3 h_3 - \rho_1 h_1$$

For small $\rho_3 h_3, \rho_1 h_1$,

$$h_2 \approx \frac{P_A - P_B}{\rho_2 \sin \theta}, \text{ as } \theta \rightarrow 0, \sin \theta \rightarrow 0, h_2 \uparrow$$

↳ Tips and tricks

1.) Same elevation + continuous volume of liquid $\rightarrow$ equal P.

2.) $P \uparrow$ as one goes down liquid; $P \downarrow$ as one goes up liquid

3.) When both liquids/gases are involved, gas effects can be neglected, $\rho_L \gg \rho_g$

$$\Delta P = \sum_{\text{down}} \rho_i g h_i - \sum_{\text{up}} \rho_i g h_i$$

↳ Hydrostatic Forces on Submerged Surfaces

$$d\vec{F}_p = -P dA \hat{n}$$

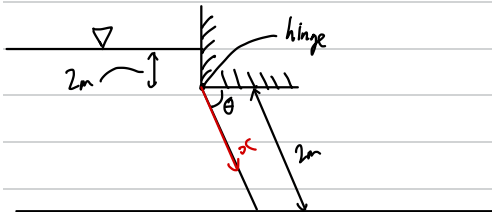
$$= -P d\vec{A}$$

$$\vec{F}_R = \int_A -P d\vec{A}$$

Resultant
Pressure Force

↳ Hydrostatic Force on Plane surface

Example question:



where $\sin \theta = 0.8$
 $\cos \theta = 0.6$
 gate width = 1.8m

Find m_{mass} of gate s.t. the gate will open when the water level $> 2\text{m}$

↳ Integration method

1.) Find closing moment



$$M = mg \cos \theta \times 1$$

$$= 0.6 mg$$

2.) Opening moment:



$$M = \int_A x P dA$$

$$= \int_A x P d\vec{A}$$

$$\sin \theta = \frac{h}{x}$$

$$P = P_0 + \rho g (2) + \rho g (h), \quad h = x \sin \theta$$

$$= \rho g (2 + x \sin \theta)$$

$$M = \int_A x \rho g (2 + x \sin \theta) dA, \quad dA = w dx$$

$$= w \int_0^2 \rho g (2x + x^2 \sin \theta) dx$$

$$= w \left(\rho g x^2 + \frac{\rho g \sin \theta x^3}{3} \right) \Big|_0^2$$

$$= 1.8 \rho g \left(4 + \frac{0.8 \times 8}{3} \right)$$

3.) $M_{\text{open}} = M_{\text{close}}$

$$0.6 mg = 1.8 \rho g \left(4 + \frac{0.8 \times 8}{3} \right)$$

$$m = 3 \left(4 + \frac{0.8 \times 8}{3} \right) \rho$$

$$= 18.4 \rho$$

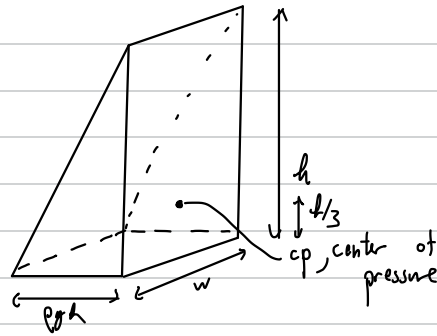
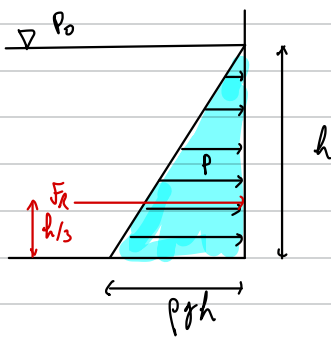
$$= 18400 \text{ kg}$$

→ Pressure Prism

- Base Case:

Planner:

3D:

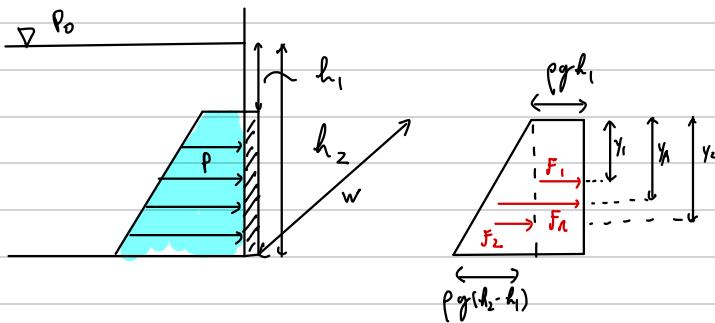


$$P_{avg} = \frac{P_{max}}{2} = \frac{P_0 + pgh}{2} = pgh/2$$

$$F_R = P_{avg} A = (pgh/2) hw = \text{Volume of pressure prism}$$

→ Note Application location = centroid = $h/3$

- Surface not exposed to fluid surface



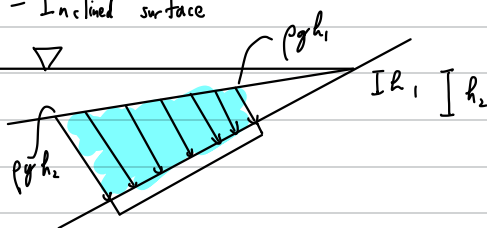
$$F_R = F_1 + F_2 \quad \text{volume of rectangular block} + \text{volume of triangular prism}$$

$$\text{And } F_R y_A = F_1 y_1 + F_2 y_2$$

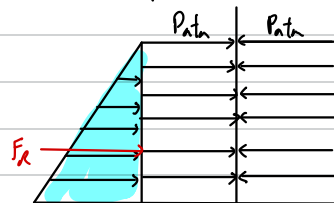
$$y_A = \frac{F_1 y_1 + F_2 y_2}{F_1 + F_2} \quad \text{sum of moments}$$

$$\text{(Here, } y_1 = \frac{h_2 - h_1}{2}, y_2 = \frac{2}{3}(h_2 - h_1) \text{)}$$

- Inclined surface

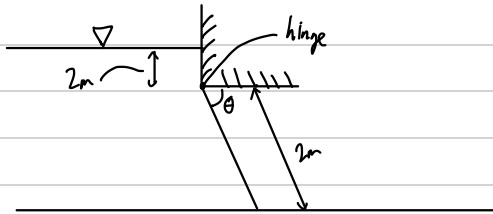


- Absolute pressure



- In cases the area is non rectangular, integration is required to find the centroid.

Pressure Prism Method:

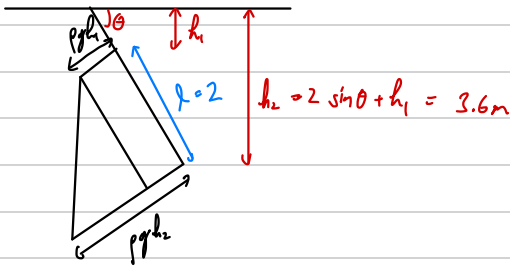


where $\sin \theta = 0.8$
 $\cos \theta = 0.6$
 gate width = 1.8m

Find m , mass of gate s.t. the gate will open when the water level $> 2m$

1.) $M = 0.6mg$

2.) Pressure prism:



$$F_R = \frac{\rho g (h_1 + h_2) l w}{2} = \rho g (h_1 + h_2) w = \rho g (10.8) = 100800N$$

$$3) F_1 = \rho g h_1 l w = 2(1.8)(2) \rho g = 72000N$$

$$F_2 = F_R - F_1 = 28800N$$

$$y_1 = \frac{l}{2} = 1m$$

$$y_2 = \frac{2l}{3} = \frac{4}{3}m$$

$$y_A = \frac{y_1 F_1 + y_2 F_2}{F_R}$$

$$= 1.0952m$$

$$M = F_R y_A = 110900$$

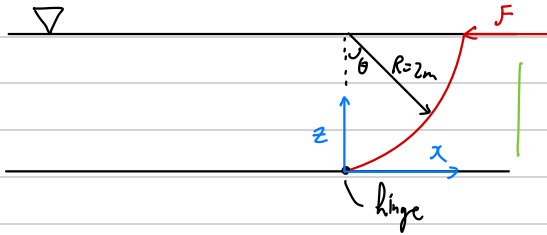
$$4) M_{open} = M_{close}$$

$$110900 = 0.6mg$$

$$m = 18400kg$$

↳ Hydrostatic Force on Curved Surface

Example question:



$$w = 6m$$

$$R = 2m$$

Find F.s.t. the gate is closed,
neglect the mass of gate

Line Integral



$$dM = x |dF_z| + z |dF_x|$$

Recall, $dF = -p d\vec{A}$

$$|dF| = P \, w \, ds$$

Then, $|dF_3| = P_w \cos \theta ds$

$$|dF_{su}| = P_w \sin \theta ds$$

$$dM = \gamma (P_w \cos \theta ds) + z (P_w \sin \theta ds)$$

$$dM = P_w (x \cos \theta + z \sin \theta) ds$$

Parameterize, $s = (x, z)$
 $= (r \sin \theta, r - r \cos \theta)$

$$\longrightarrow M = w \int_0^{\pi/2} \rho (x \cos \theta + z \sin \theta) r d\theta$$

$$M = \omega \int_0^{\pi/2} P (\cos \theta \sin \theta \cos \theta + r \sin \theta - r \sin \theta \cos \theta) r d\theta$$

$$M = w^{r/2} \int_0^{r/2} P r^2 \sin \theta \, d\theta$$

$$P = \rho g h, \quad h = r \cos \theta$$

$$M = w r^3 \rho y_0 \int_0^{\pi/2} \sin \theta \cos \theta d\theta$$

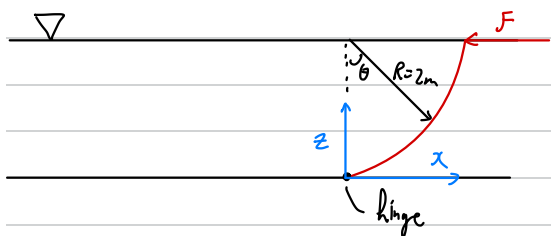
$$= \frac{\omega^2 \rho g}{4}$$

$$= 240,000 \text{ N}$$

Thus $M_{eq} = 240,000 \text{ kg m}^2/\text{s}^2$

$$F_r = 240,000 \text{ N/m}$$

$$F = 120,000 \text{ N}$$

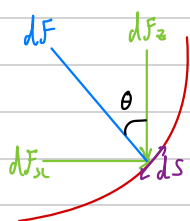


$$w = 6m$$

$$R = 2m$$

Find F s.t. the gate is closed,
neglect the mass of gate

Surface
Integral



Parametrize curve:

$$\vec{r}(\theta, y) = r \sin \theta \hat{i} + y \hat{j} + r(1 - \cos \theta) \hat{k}$$

$$\vec{r}_\theta = r \cos \theta \hat{i} + 0 \hat{j} + r \sin \theta \hat{k}$$

$$\vec{r}_y = \hat{j}$$

$$\vec{r}_\theta \times \vec{r}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r \cos \theta & 0 & r \sin \theta \\ 0 & 1 & 0 \end{vmatrix} = -r \sin \theta \hat{i} + r \cos \theta \hat{k}$$

$$d\vec{F}_p = -P \hat{n} dS$$

$$= -P \left(\frac{\vec{r}_\theta \times \vec{r}_y}{|\vec{r}_\theta \times \vec{r}_y|} \right) |\vec{r}_\theta \times \vec{r}_y| d\theta dy$$

$$d\vec{F}_p = (P r \sin \theta \hat{i} - P r \cos \theta \hat{k}) d\theta dy \rightarrow \text{positive } x \text{-direction}$$

$$|d\vec{F}_x| = P r \sin \theta d\theta dy$$

$$|d\vec{F}_z| = P r \cos \theta d\theta dy$$

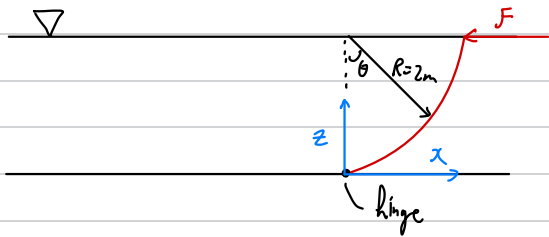
$$dU = |d\vec{F}_x|_x + |d\vec{F}_z|_z$$

$$= P r \sin \theta (r - r \cos \theta) + P r \cos \theta r \sin \theta d\theta dy$$

$$= P r^2 \sin \theta d\theta dy$$

$$\int_0^w \int_0^{\pi/2} P r^2 \sin \theta d\theta dy$$

$$= \int_0^w \int_0^{\pi/2} \rho g r^3 \cos \theta \sin \theta d\theta dy$$



$$w = 6m$$

$$R = 2m$$

Find F s.t. the gate is closed,
neglect the mass of gate

Parametrize curve:

$$\vec{r}(\theta, y) = r \sin \theta \hat{i} + y \hat{j} + r(1 - \cos \theta) \hat{h}$$

$$\vec{r}_\theta = r \cos \theta \hat{i} + 0 \hat{j} + r \sin \theta \hat{h}$$

$$\vec{r}_y = \hat{j}$$

$$\vec{r}_\theta \times \vec{r}_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{h} \\ r \cos \theta & 0 & r \sin \theta \\ 0 & 1 & 0 \end{vmatrix} = -r \sin \theta \hat{i} + r \cos \theta \hat{h}$$

$$d\vec{F} = -P \vec{n} dS$$

$$= -P (-r \sin \theta \hat{i} + r \cos \theta \hat{h}) d\theta dy$$

$$= P r \sin \theta \hat{i} - P r \cos \theta \hat{h} d\theta dy$$

$$\vec{r} \times d\vec{F}$$

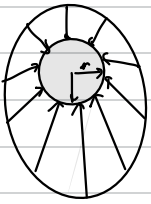
$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{h} \\ r \sin \theta & y & r(1 - \cos \theta) \\ P r \sin \theta & 0 & -P r \cos \theta \end{vmatrix} = \dots \hat{i} - (-P r^2 \sin \theta \cos \theta - P r^2 \sin \theta (1 - \cos \theta)) \hat{j} + \dots \hat{h} d\theta dy$$

$$dM_y = (\vec{r} \times d\vec{F}) \cdot \hat{j} = P r^2 \sin \theta d\theta dy$$

$$M_y = \dots$$

↳ FBD Method.

↳ Buoyancy Force



$$d\vec{F} = -P \cdot \hat{n} dA$$

$$\text{let } \vec{r}(\theta, y) = r \cos \theta \hat{i} + y \hat{j} + r \sin \theta \hat{k}$$

$$r_{\theta} = -r \sin \theta \hat{i} + r \cos \theta \hat{k}$$

$$r_y = \hat{j}$$

$$\vec{n} = r_{\theta} \times r_y = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ r \cos \theta & y & r \sin \theta \\ -r \sin \theta & 0 & r \cos \theta \end{vmatrix}$$

$$= y r \cos \theta \hat{i} - (r^2 \cos^2 \theta + r^2 \sin^2 \theta) \hat{j} + y r \sin \theta \hat{k}$$

$$= (y r \cos \theta, -r^2, y r \sin \theta)$$

$$r_y \times r_{\theta} = (-y r \cos \theta, r^2, -y r \sin \theta)$$

$$P = P_0 + \rho g r \sin \theta$$

$$F = \iint -P \vec{n} d\theta dy$$

$$= \int_0^l \int_0^{2\pi} (P_0 + \rho g r \sin \theta) (-y r \cos \theta, 0, -y r \sin \theta) d\theta dy$$

$$= (0, 0, l \int_0^{2\pi} \rho g r^2 \sin^2 \theta d\theta)$$

$$= (0, 0, l \rho g r^2)$$

$$= (0, 0, \rho g V)$$

↳ Archimedes' Principle

- Buoyant force, F_B , for body immersed in fluid = weight of fluid displaced by body, acting on the **centroid of displaced volume**

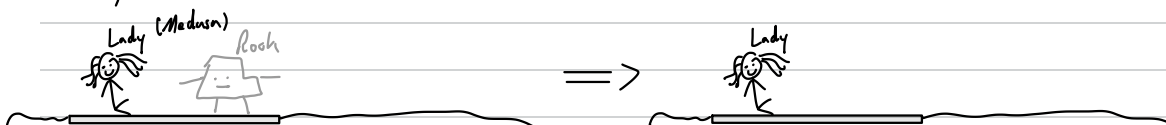
- For floating bodies, $F_B = W$

$$F_B = W$$

$$\rho_w g V_w = \rho_s g V_s$$

$$\frac{\rho_w}{\rho_s} = \frac{V_s}{V_w}$$

↳ Lady and Rock example



Only consider buoyancy due to stone:



$$\rho_f g V_f = \rho_s g V_s$$

$$\frac{\rho_f}{\rho_s} = \frac{V_s}{V_f}$$

$$\rho_s > \rho_f$$

$$\text{thus } \frac{V_s}{V_f} < 1$$

$$V_s < V_f$$

Volume of fluid displaced >
Volume of rock

Now, when rock sinks,

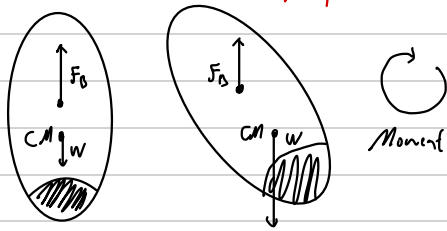
$$V_f' = V_s$$

$$V_f' < V_f //$$

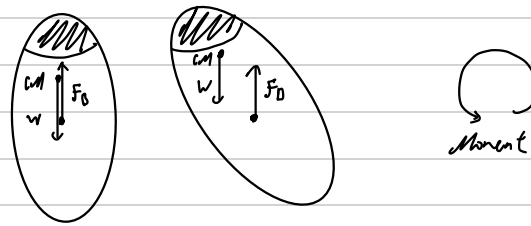
↳ Stability of immersed, floating objects

For completely submerged body:

CM below center of buoyancy:



CM above center of buoyancy:



↳ Fluid in linear motion

$$-\vec{\nabla} p - \rho g \hat{h} = \rho \vec{a}$$

$$-\left(\frac{\partial p}{\partial x} \hat{i} + \frac{\partial p}{\partial y} \hat{j} + \frac{\partial p}{\partial z} \hat{h}\right) - \rho g \hat{h} = \rho \vec{a}$$

$$\hat{j}: \frac{\partial p}{\partial y} = -\rho a_y$$

$$\hat{h}: \frac{\partial p}{\partial z} = -\rho(a_z + g)$$

$$\frac{\partial p}{\partial z} = -\rho(a_z + g)$$

$$dP = \frac{\partial P}{\partial y} dy + \frac{\partial P}{\partial z} dz$$

$$= -\rho a_y dy - \rho(a_z + g) dz$$

$$P = -\rho a_y y - \rho(a_z + g)z + C$$

$$@ z = z_s, P = P_{atm}$$

$$P_{atm} = -\rho(a_y y + (a_z + g)z_s) + C$$

$$P_{atm} = \rho(a_y y + (a_z + g)z_s)$$

$$\rho(a_y y + (a_z + g)z_s) = C - P_{atm}$$

$$(a_z + g)z_s = \frac{C - P_{atm}}{\rho} - a_y y$$

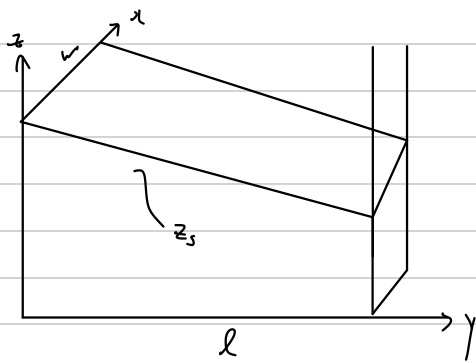
$$z_s = \frac{C - P_{atm}}{\rho(a_z + g)} - \frac{a_y y}{(a_z + g)}$$

equation of surface, slope = $\frac{-a_y}{a_z + g}$

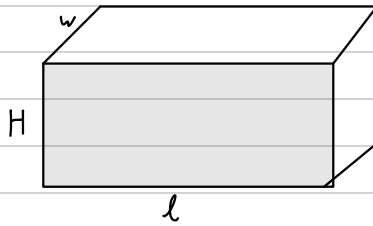
$$(z = ay + b)$$

Equation Volumes

$$z_s = \frac{-a_y y}{(a_z + g)}$$



At rest!



$$\int_0^l z_s dy = H l$$

$$H l - \frac{a_y l^2}{2(a_z + g)} = H l$$

$$H = H + \frac{a_y l}{2(a_z + g)}$$

$$z_s = H + \frac{a_y l}{2(a_z + g)} - \frac{a_y y}{(a_z + g)}$$

$$= H + \frac{a_y}{a_z + g} \left(-y + \frac{l}{2} \right)$$

Special cases

- $a_z = 0$, acceleration along straight line, in y-direction
 $\rightarrow$ free surface inclined for $a_y \neq 0$

- $a_y = 0$, $z_s = H$ or

$$\frac{dP}{dz} = -\rho(a_z + g) \quad \text{and} \quad \frac{dP}{dy} = 0$$

$$P = -\rho(a_z + g)z \leftarrow \text{indep of } y$$

- Free falling: $a_x = 0, a_y = 0, a_z = -g$

$$\frac{dP}{dz} = 0 = \frac{\partial P}{\partial x} = \frac{\partial P}{\partial y} \rightarrow P \text{ is const.}$$

Rigid body rotation

$$a = -\frac{v^2}{r}, v = \omega r \rightarrow a = -\omega^2 r$$

$$-\vec{\nabla} P - \rho g \hat{n} = \rho \vec{a}$$

$$\vec{\nabla} P = \frac{\partial P}{\partial r} \hat{e}_r + \frac{1}{r} \frac{\partial P}{\partial \theta} \hat{e}_\theta + \frac{\partial P}{\partial z} \hat{n}$$

$$\vec{a} = -\omega^2 r \hat{e}_r + 0 \hat{e}_\theta + 0 \hat{n}$$

$$\frac{\partial P}{\partial r} = -\rho \omega^2 r, \frac{1}{r} \frac{\partial P}{\partial \theta} = 0, \frac{\partial P}{\partial z} = \rho g$$

$$dP = \frac{\partial P}{\partial r} dr + \frac{\partial P}{\partial z} dz$$

$$= -\rho \omega^2 r dr + \rho g dz$$

$$P = -\frac{\rho \omega^2 r^2}{2} + \rho g z + C$$

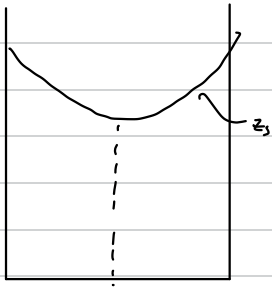
$$\text{at } z_s, P = P_{atm}$$

$$P_{atm} = -\frac{\rho \omega^2 r^2}{2} + \rho g z_s + C$$

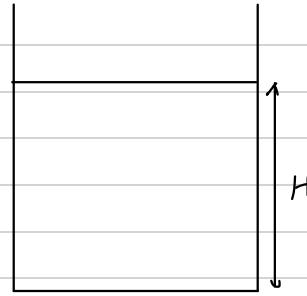
$$P_{atm} + \frac{\rho \omega^2 r^2}{2} - C = \rho g z_s$$

$$z_s = \frac{P_{atm} - C}{\rho g} + \frac{\omega^2 r^2}{2g}$$

$$z_s = h + \frac{\omega^2 r^2}{2g}$$



$$\begin{aligned}
 V &= \int_0^{2\pi} \int_0^R z_s r dr d\theta \\
 &= 2\pi \int_0^R \left(kr + \frac{\omega^2 r^3}{2g} \right) dr d\theta \\
 &= 2\pi \int_0^R \left(\frac{kR^2}{2} + \frac{\omega^2 R^4}{8g} \right) d\theta \\
 &= 2\pi \left(\frac{kR^2}{2} + \frac{\omega^2 R^4}{8g} \right) \\
 &= \pi R^2 \left(k + \frac{\omega^2 R^2}{4g} \right)
 \end{aligned}$$



$$V = \pi R^2 H$$

$$H = k + \frac{\omega^2 R^2}{4g}$$

$$k = H - \frac{\omega^2 R^2}{4g}$$

$$\text{and } z_s = H - \frac{\omega^2 R^2}{4g} + \frac{\omega^2 r^2}{2g}$$

$$= \frac{\omega^2}{4g} (2r^2 - R^2) + H$$

↳ Conclusion

- Linear Motion: $z_s = H + \frac{a_z}{g} (-y + \frac{L}{2})$, linear line
- Rigid rotation: $z_s = H + \frac{\omega^2}{4g} (2r^2 - R^2)$, parabola

4 Flowing Fluids

↳ Lagrangian approach

- Identifying a small mass of fluid (fluid particle)
- Describe motion with time

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}$$

$$\vec{v}(t) = x'(t)\hat{i} + y'(t)\hat{j} + z'(t)\hat{k}$$

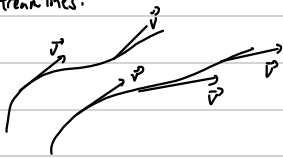
↳ Eulerian approach

- Information of velocity of fluid particles that pass each window for all time

$$\vec{v}(t) = \vec{v}(x, y, z, t) = \vec{v}(x_0, y_0, z_0, t)$$

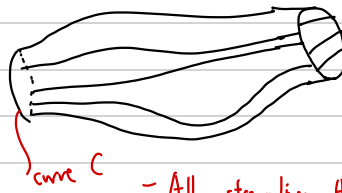
↳ Flow visualization

Streamlines:



- ↳ Line that is tangent to local velocity vector at every point along the line at that instant

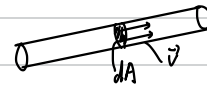
Streamtubes:



- All streamlines that pass through all points on arbitrary closed curve C

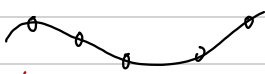
- NO fluid within a streamtube can pass through the boundary of the streamtube

Stream filament



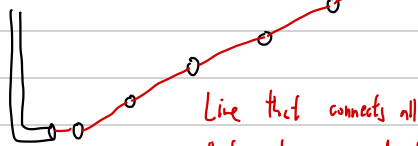
- Small enough streamtube s.t. velocity is constant over dA

Pathlines



Actual path the particle takes

Streaklines



Line that connects all fluid particles that have passed through the same point in space at a previous time

↳ Steady / Unsteady flows

- Steady

- o Flow properties do not depend on time (unsteady if otherwise)
- o They can be a function of space
- o Streamline, streakline, pathline are equal (unsteady otherwise)

↳ Dimensionality

- One-, two-, three-dimensional if flow velocity / properties vary with one-, two-, three- spatial dimensions.
- Depends on coordinate system chosen

↳ Viscous / Inviscid

- Viscous:

Frictional effects / shear force is significant

- Inviscid:

Viscous forces are negligibly small compared to inertial / pressure forces

↳ Laminar / Turbulent

- Laminar: smooth, orderly $\rightarrow$ high viscosity, low velocity
- Turbulent: Chaotic $\rightarrow$ low viscosity, high velocity
- Transitional: In b/w laminar, turbulent

Higher $Re = \frac{\rho V L}{\mu} \Rightarrow$ Turbulent flow

↳ Volume rate of flow

$$\dot{V} = \frac{dV}{dt} \quad \text{and} \quad dV = \vec{V} \cdot d\vec{A} \, dt$$

$$\dot{V} = \iint_A \vec{V} \cdot d\vec{A}$$

$$= \iint_A \vec{V} \cdot \vec{n} \, dA$$

$$= \iint_A V_n \, dA$$

normal component

$$V_n = V \cos \theta$$

$$= V_{n, \text{avg}} A$$

average of normals



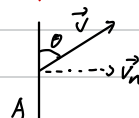
↳ Mass flow rate

$$\dot{m} = \iint_A \rho \vec{V} \cdot d\vec{A}$$

for uniform velocity, density,

$$\dot{m} = \rho V_n A$$

$$= \rho V A \cos \theta$$



for uniform distribution,

$$\vec{V} = V_{avg} \vec{n}, \quad \dot{V} = V_n A$$

↳ Conservation of Mass

For steady, incompressible flow,

$$\dot{m}_1 = \dot{m}_2$$

$$\iint \rho_1 V_1 \, dA_1 = \iint \rho_2 V_2 \, dA_2 \quad \text{but} \quad \rho_1 = \rho_2$$

$$\iint V_1 \, dA_1 = \iint V_2 \, dA_2$$

$$V_{avg,1} A_1 = V_{avg,2} A_2 \quad (\text{Steady, incompressible})$$

$$V_1 A_1 = V_2 A_2 \quad (\text{1D flow case, uniform velocity})$$

For steady, compressible flow,

$$\iint \rho_1 V_1 \, dA_1 = \iint \rho_2 V_2 \, dA_2$$

For 1D case,

$$\rho_1 V_1 A_1 = \rho_2 V_2 A_2$$

↳ Conservation of energy

- Euler's Equation

$$V dV + g dz + \frac{1}{\rho} dP = 0$$

- Steady

- Non-viscous

- Along a streamline

★ - Compressible

- Bernoulli's Equation

$$\frac{V^2}{2} + gz + \frac{P}{\rho} = C$$

- Steady

- Non-viscous

- Along a streamline

★ - Incompressible

Additionally,

- No heat exchange

- No shaft work

$\rightarrow$ Flow along straight line:

$$P_1 - P_2 = \rho g (z_2 - z_1)$$

Energy form:

$$\frac{P}{\rho} + \frac{V^2}{2} + gz = E_T$$

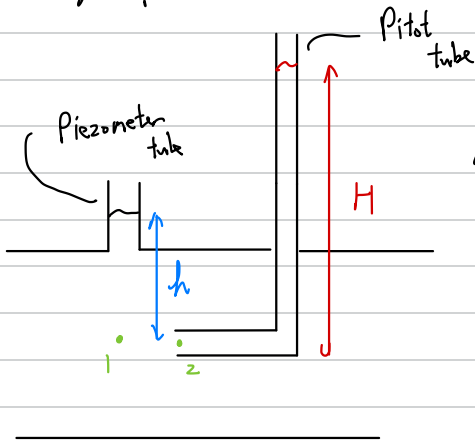
Energy due to pressure / unit mass Kinetic energy per unit mass Potential Energy per unit mass

$$P + \frac{\rho V^2}{2} + \rho g z = \text{constant} \quad \text{Total P}$$

static pressure Dynamic pressure Hydrostatic pressure

Actual thermodynamic pressure of fluid Pressure rise when fluid is brought to stop isentropically Accounts for elevation effects

↳ Measuring pressure



a.) Static Pressure = $\rho g h$

b.) Apply Bernoulli,

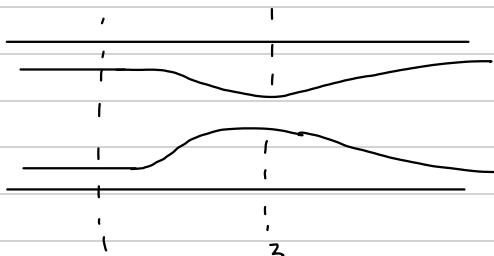
$$P_2 = P_1 + \frac{1}{2} \rho V_1^2$$

stagnation pressure static pressure dynamic pressure

↳ Torricelli equation

$$V = \sqrt{2gz}$$

↳ Measuring flow rate



a.) Bernoulli:

$$\frac{P_1}{\rho} + \frac{V_1^2}{2} + gz_1 = \frac{P_2}{\rho} + \frac{V_2^2}{2} \rightarrow \frac{V_2^2 - V_1^2}{2} = \frac{P_1 - P_2}{\rho}$$

b.) Continuity:

$$P_1 A_1 V_1 = P_2 A_2 V_2$$

$$A_1 V_1 = A_2 V_2$$

$$\text{Sub } V_1 = \frac{A_2}{A_1} V_2$$

$$V_2^2 \left(1 - \frac{A_2^2}{A_1^2} \right) = \frac{2}{\rho} (P_1 - P_2)$$

c.) $\dot{V} = A_2 V_2$

$$V_2^2 = \frac{2(P_1 - P_2)}{(1 - A_2^2/A_1^2)}$$

$$\dot{V} = A_2 \sqrt{\frac{2(P_1 - P_2)}{(1 - A_2^2/A_1^2)}}$$

$$V_2 = \sqrt{\frac{2(P_1 - P_2)}{(1 - A_2^2/A_1^2)}}$$

↳ Control volume approach

- Volume in the field is identified, governing equations are solved for the properties associated with the volume
- **Eulerian concept**
- Control volume = "window"

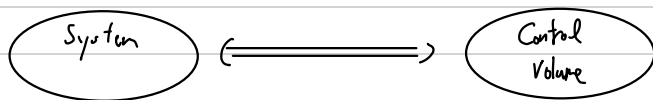
↳ System approach

- Cont. collection of mass of fluid system that moves with the flow, solving equation for the system.
- **Lagrangian concept**

↳ Definitions

- Control volume, volume in space
- Control surface, surface that encloses control volume.
 - o Mass can cross the surface, it can be changing
- Fluid system, cont. mass of fluid that always contains the same fluid particles
 - o Mass is constant

↳ Reynolds Transport Theorem



- Parameter/unit mass:

$$B = m b$$

For system: dm

$$B_{sys} = \int_{sys} b \rho dV$$

For control volume:

$$B_{control} = \int_{cv} b \rho dV$$

over control volume

Rate of change:

$$\frac{dB_{sys}}{dt} = \frac{d}{dt} \int_{sys} b \rho dV$$

$$\frac{dB_{cv}}{dt} = \frac{d}{dt} \int_{cv} b \rho dV$$

Derivation:

$$B_{sys}(t) = B_{cv}(t)$$

$$B_{sys}(t + \Delta t) = B_{cv}(t + \Delta t) + B_{out} - B_{in}$$

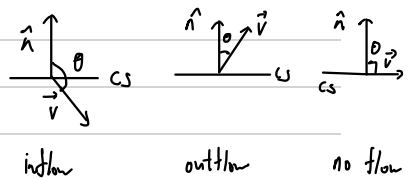
$$\begin{aligned} \frac{dB_{sys}}{dt} &= \lim_{\Delta t \rightarrow 0} \frac{B_{sys}(t + \Delta t) - B_{sys}(t)}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{B_{cv}(t + \Delta t) + B_{out} - B_{in} - \overbrace{B_{sys}(t)}^{B_{cv}(t)}}}{\Delta t} \\ &= \lim_{\Delta t \rightarrow 0} \frac{B_{cv}(t + \Delta t) - B_{cv}(t)}{\Delta t} + \frac{B_{out}}{\Delta t} - \frac{B_{in}}{\Delta t} \\ &= \frac{dB_{cv}}{dt} + \dot{B}_{out} - \dot{B}_{in} \end{aligned}$$

$$\dot{B} = \dot{B}_{\Delta t}$$

Generally, $\frac{dB_{sys}}{dt} = \frac{dB_{cv}}{dt} + \underbrace{\oint_{cs} b \rho (\vec{V} \cdot d\vec{A})}_{\substack{\text{time rate of net flow of } B \\ \text{from control volume through} \\ \text{control surface}}}$ $\xrightarrow{\text{net outflow}} \dot{B}_{out} - \dot{B}_{in}$

$$-\vec{V} \cdot d\vec{A} = \|\vec{V}\| \|d\vec{A}\| \cos \theta$$

$$\theta > \pi/2: \quad \theta < \pi/2: \quad \theta = \pi/2$$



↳ Continuity from RTT

$$\frac{dB_{sys}}{dt} = \frac{dB_{cv}}{dt} + \oint_{cs} b \rho (\vec{V} \cdot d\vec{A})$$

$$B = m, \quad b = 1, \quad \frac{dB_{sys}}{dt} = 0 //$$

$$\frac{dm_{cv}}{dt} + \oint_{cs} \rho (\vec{V} \cdot d\vec{A}) = 0$$

$$\frac{d}{dt} \underbrace{\iiint_V \rho dV}_{m_{cv}} + \oint_{cs} \rho (\vec{V} \cdot d\vec{A}) = 0$$

$$(1) \text{ Steady flow, } \frac{d}{dt} \iiint_V \rho dV = 0 \longrightarrow \oint_{cs} \rho (\vec{V} \cdot d\vec{A}) = 0 \text{ or } \iint_{A_1} \rho_1 \vec{v}_1 dA_1 = \iint_{A_2} \rho_2 \vec{v}_2 dA_2$$

$$(2) \text{ Steady, Incompressible, } \rho \iint_{A_1} \vec{v}_1 dA_1 = \rho \iint_{A_2} \vec{v}_2 dA_2$$

$$(3) \text{ Steady, 1D, incompressible, } v_1 A_1 = v_2 A_2$$

$\rightarrow$ constant properties over cross section

$$(4) \text{ Steady, 1D, compressible, } v_1 \rho_1 A_1 = v_2 \rho_2 A_2$$

↳ Momentum Principle from RTT

$$\frac{d\vec{B}_{sys}}{dt} = \frac{d\vec{B}_{cv}}{dt} + \oint \rho \vec{v} (\vec{v} \cdot d\vec{A})$$

$$\vec{B} = m\vec{v}$$

$$\vec{B} = \vec{v}$$

$$\frac{dm\vec{v}_{sys}}{dt} = \frac{dm\vec{v}_{cv}}{dt} + \dot{M}_{net} \quad \text{or} \quad \oint \rho \vec{v} (\vec{v} \cdot d\vec{A})$$

① Using Lagrangian physics, ② And at time t , $\sum \vec{F}_{sys} = \sum \vec{F}_{cv}$

$$\sum \vec{F}_{sys} = \frac{d(m\vec{v}_{sys})}{dt}$$

$$\sum \vec{F}_{cv} = \frac{d(m\vec{v}_{cv})}{dt} + \dot{M}_{net}$$

$$\sum \vec{F}_{cv} = \frac{d}{dt} \iiint \rho \vec{v} dV + \oint \rho \vec{v} (\vec{v} \cdot d\vec{A})$$

Special cases:

① Steady state, $\sum \vec{F}_{cv} = \oint \rho \vec{v} (\vec{v} \cdot d\vec{A})$, $\frac{d}{dt} \iiint \rho \vec{v} dV = 0$

② Steady, 1D flow,

$$\sum \vec{F}_{cv} = \sum \dot{m}_{out} \vec{v}_{out} - \sum \dot{m}_{in} \vec{v}_{in}$$

③ 1 inlet, 1 outlet,

5. General Forms of Continuity

Substantial Derivative:

$$\frac{D}{Dt} \equiv \frac{\partial}{\partial t} + \frac{dx}{dt} \cdot \frac{\partial}{\partial x} + \frac{dy}{dt} \frac{\partial}{\partial y} + \frac{dz}{dt} \frac{\partial}{\partial z}$$

Substantial: Time rate of change following a moving infinitesimal system

Local: Time rate of change at a fixed point due to unsteady fluctuations of that property

Convective: Time rate of change due to movement of fluid element from 1 location to another in the flow field, the properties are spatially different

Density:

$$\frac{D\rho}{Dt} = \frac{\partial \rho}{\partial t} + (\vec{v} \cdot \vec{\nabla})\rho$$

Divergence of Velocity

$$\vec{\nabla} \cdot \vec{v} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = \frac{1}{\delta V} \frac{D\delta V}{Dt}$$

Time rate of change of volume of an infinitesimal moving system / unit volume

4 Continuity Eqs

Finite CV
fixed in space

$$\frac{\partial}{\partial t} \iiint_V \rho \, dV + \oint_S \rho \vec{v} \cdot d\vec{S} = 0$$



$$\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v}) = 0$$

Infinitesimal CV
fixed in space

Finite system with
constant mass moving with flow

$$\frac{D}{Dt} \iiint_V \rho \, dV = 0$$



$$\frac{D\rho}{Dt} + \rho(\vec{\nabla} \cdot \vec{v}) = 0$$

Infinitesimal system of
constant mass moving with the flow

Steady var

Finite CV
fixed in space

$$\cancel{\frac{d}{dt} \iiint_V \rho \, dV} + \oint_S \rho \vec{v} \cdot d\vec{S} = 0$$



$$\cancel{\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v})} = 0$$

Infinitesimal CV
fixed in space

Finite system with
constant mass moving with flow

$$\frac{D}{Dt} \iiint_V \rho \, dV = 0$$



$$\frac{D\rho}{Dt} + \rho(\vec{\nabla} \cdot \vec{v}) = 0$$

Infinitesimal system of
constant mass moving with the flow

Note $\frac{D\rho}{Dt} \neq 0$

Incompressible var

Finite CV
fixed in space

$$\cancel{\frac{d}{dt} \iiint_V \rho \, dV} + \oint_S \rho \vec{v} \cdot d\vec{S} = 0$$



$$\cancel{\frac{\partial \rho}{\partial t} + \vec{\nabla} \cdot (\rho \vec{v})} = 0$$

Infinitesimal CV
fixed in space

Finite system with
constant mass moving with flow

$$\frac{D}{Dt} \iiint_V \rho \, dV = 0$$

$$\frac{D\rho}{Dt} = 0$$



$$\vec{\nabla} \cdot \vec{v} = 0$$

$$\frac{D\rho}{Dt} + \rho(\vec{\nabla} \cdot \vec{v}) = 0$$

Infinitesimal system of
constant mass moving with the flow

$$\oint_S \vec{v} \cdot d\vec{S} = 0$$

$$\vec{\nabla} \cdot \vec{v} = 0$$

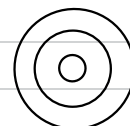
6 Open Channel Flows

Wave speed

$$c_0 \approx \sqrt{gy} \quad \text{for } \delta y \ll y$$

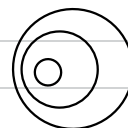
(wave speed) $\nearrow$ depth

$Fr = 0$ still fluid



Froude Number

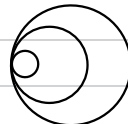
$Fr < 1 \rightarrow$ subcritical



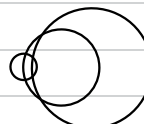
$$Fr = \frac{V}{\sqrt{gy}}$$

$\nearrow$ fluid / flow speed
 $\nwarrow$ wave speed

$Fr = 1 \rightarrow$ critical



$Fr > 1 \rightarrow$ supercritical



7. Compressible flows

Speed of sound: $c^2 = \left. \frac{dP}{d\rho} \right|_{\text{const } s}$

Ideal gas: $P = \rho R T$, isentropic condition: $\frac{P}{\rho^\gamma} = \text{const.}$

$$\hookrightarrow c = \sqrt{\gamma R T}$$

For any fluids:

$$E_v = \frac{dP}{d\rho} \rightarrow \frac{dP}{d\rho} = \frac{E_v}{\rho}$$

$$c = \sqrt{\frac{E_v}{\rho}}$$

$$E_v \rightarrow \infty \rightarrow c \rightarrow \infty$$

Mach Number

$$M = \frac{V}{c} \quad \begin{array}{l} \sim \text{flow velocity, (or velocity of object in still fluid)} \\ \sim \text{speed of sound} \end{array}$$

Compressibility: $\begin{cases} M_\infty \leq 0.3 \rightarrow \text{incompressible} \\ M_\infty > 0.3 \rightarrow \text{compressible} \end{cases}$

$$\begin{array}{ll} M_\infty < 1 & \text{Subsonic} \\ M_\infty = 1 & \text{sonic} \\ M > 1 & \text{Supersonic} \end{array}$$

$$\begin{array}{ll} 0.8 \leq M_\infty < 1 & \text{Transonic (Phases)} \\ M_\infty \geq 1 & \text{Hypersonic (Spacecraft entering atmosphere)} \end{array}$$

Steady Isentropic Flow of compressible flows

$$h_2 + \frac{V_2^2}{2} + g z_2 = h_1 + \frac{V_1^2}{2} + g z_1$$

Stagnation enthalpy:

$$h + \frac{V^2}{2} + g z^0 = \text{const}$$

if we immediately stop the obj

$$h_2 + \frac{V_2^2}{2} \xrightarrow{\text{stop}} h_1 + \frac{V_1^2}{2} = h_0 \text{ (stagnation enthalpy)}$$

$$hE \rightarrow h$$

Stagnation Temp:

$$c_p (T - T_0) + \frac{V^2}{2} = 0$$

$$T_0 = T + \frac{V^2}{2 c_p}$$

$\uparrow$ stagnation Temp $\uparrow$ static temp $\uparrow$ dynamic temp

$T_0 \approx T$ for low speed flow

Fluid Velocity with area

$$\frac{dA}{A} = -\frac{dV}{V} (1-M^2)$$

Subsonic flow ($M < 1$)

$A \uparrow \rightarrow V \downarrow$

$A \downarrow \rightarrow V \uparrow$

sonic flow ($M = 1$)

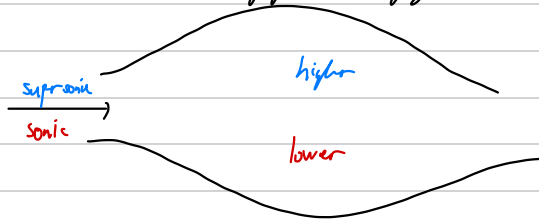
$A \rightarrow A_{min} / A_{throat}$

supersonic flow ($M > 1$)

$A \uparrow \rightarrow V \uparrow$

$A \downarrow \rightarrow V \downarrow$

Diverging - Converging duct



Converging - Diverging duct

